Microscopic Quantum Mechanisms of Stabilization of Relativistic Electro-Weak Objects with Critical Mass 2.8 Msun - Core Structure Hypothesis

Microscopic Quantum Mechanisms of Stabilization of Relativistic Electro-Weak Objects with Critical Mass 2.8 Msun. Core, core (Black hole?).

Marcin Slowik - Author's hypothesis
ORCID: 0009-0004-6707-4546
Location: Trasa Polnocna, Zielona Gora, Poland
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(INTRODUCTION) Introduction in thought.
Imagine you take the largest star you can imagine, 30-100+ times heavier than the Sun.
Our Sun (a star with mass 1 Msun) shines thanks to nuclear fusion.
Inside it, 4 hydrogen nuclei (4 protons) combine in the proton-proton (p-p) cycle into one helium-4 nucleus (2 protons + 2 neutrons).
The helium nucleus weighs about 0.7% less than the sum of 4 protons.
This mass difference does not disappear, but transforms into energy according to E=mc2, (0.7) of the remaining mass turns into energy in the form of photons and heat.
Thanks to this we have light and heat.

QUANTUM TUNNELING (IGNITION) - (Sun)(Star)
Four protons combine into helium, releasing energy. However there is one problem: --protons have a positive charge and repel each other--.
It's like trying to bring together two magnets with the same poles - the closer, the stronger they repel.
In the core of the Sun (star) the temperature is about 15 million degrees Celsius (particle motion).
This is enormously much, but according to classical physics (our everyday physics) it is still too little for protons to overcome mutual repulsion.
If only classical physics worked, the Sun would never ignite - it would be a cold, dark gas balloon.
And here quantum mechanics enters -- the phenomenon of tunneling.
Imagine that you stand before a high, smooth wall - so high that you have no chance to jump over it or break through. Classical physics says: "Without a ladder you will not pass".
Quantum mechanics says something surprising: "You have a small, but non-zero chance that you will simply find yourself on the other side, even without going through the wall".
This is not going through the wall - it is penetrating to the other side, as if the wall in a sense ceased to exist for that particular particle.
For a single proton in the Sun, the chance of tunneling through the Coulomb barrier is improbably small - about 1e-20 (i.e. 1 to 100 billion billions).
BUT - in the Sun there are so many collisions per second (1e38) that even such a tiny probability is enough for billions of fusion reactions per second to occur.

Tunneling creates a "recoil package"?
On the Sun there are also other mechanisms:
When two protons as a result of tunneling approach to a distance at which the strong interaction (nuclear attraction) starts to act they combine into a heavier nucleus - deuterium (proton + neutron).
This process releases a specific energy package (1.44 MeV in the form of positron and neutrino).
Then deuterium combines with another proton, creating helium-3, and helium-3 combines with a second helium-3, creating helium-4 (ejecting 2 protons) releasing the main energy package 26.7 MeV.
Each such package is a small "shot" - a recoil that pushes matter outward.
Millions of billions of such packages per second create radiation pressure and gas pressure that balances gravity.
Without tunneling there is no fusion, without fusion there are no packages, without packages there is no pressure - and without pressure gravity wins and the star collapses.
This tunneling is therefore the "spark plug" for the entire mechanism of stellar equilibrium.
When hydrogen runs out in a star, it begins to fuse heavier and heavier elements, up to iron.
Fusion of iron no longer gives energy profit, but takes it. The star then loses the source of pressure that balanced gravity.
What happens next depends on the mass of the remaining core:
- If the core has mass approx. 8-20 Msun, after ejecting the envelope in a supernova explosion a

neutron star remains with mass approx. 1.4-2.0 Msun.
- If the initial star had 20-40+ Msun (hypernova), at first a massive neutron star can be formed, up to the limit of approx. 2.6 Msun.
- Above the Tolman-Oppenheimer-Volkoff (TOV) limit the core collapses further into singularity, i.e. black hole?.
(Msun) - means the mass of our Sun = 1.989 x 1e30 kg.
The remaining core usually has a radius of only 10-13 km.
Then a (Neutron Star) is formed - an object with a diameter of only 20-25 km, but so dense that a teaspoon of its matter would weigh as much as all humanity.

PART 1 - WHAT WE ALREADY KNOW.
When a massive star with mass of order 25-30 (Msun) exhausts nuclear fuel, it is no longer able to maintain equilibrium between internal pressure and its own gravity. Gravitational collapse occurs - a violent collapse of the star's core to small dimensions.
Birth of (Neutron Star)
When the core of a massive star collapses, the pressure inside rises to such an extent that electrons (leptons) are literally pushed into atomic nuclei.
Electrons and protons combine in a process called electron capture, creating neutrons and releasing electron neutrinos.
Imagine you squeeze an orange so hard that the juice (electrons) penetrates into the peel (nucleus) and changes its structure. Neutrons are created - neutral particles that can be arranged very densely.
An object with mass up to approx. 2.6 Msun, density of order 1e17-1e18 kg/m3 (comparable to atomic nucleus density) and radius of only 10-13 km:
Threshold mass 2.6 Msun Density 1e17-1e18 kg/m3 Radius 10-13 km Diameter 20-25 km
(To feel this density: a teaspoon of matter from a neutron star would weigh as much as all cars on Earth combined.)
Preventing further collapse of the neutron star.
Degeneracy of Fermi gas
A neutron star does not collapse further thanks to a phenomenon called degeneracy of fermionic gas (fermions - quarks, leptons). It is a state of matter in which pressure does not result from temperature but directly from the laws of quantum mechanics.
It's like trying to put 100 people into a passenger car. Even if everyone is calm, there is simply no space. Particles also have their "place" they cannot be in the same quantum state.
Pauli Exclusion Principle
Prevents two neutrons (fermions) from occupying the same quantum state.
When particles are packed extremely tightly, the uncertainty principle forces an enormous spread of momenta on them. This forced motion of particles generates degeneracy pressure - independent of temperature.
(Even if the neutron star were completely cold, particles inside would still "dance" - but this dance is not heat, only quantum compulsion resulting from the fact that they cannot be in the same place in the same way. This pressure prevents the star from complete collapse.)
The maintenance of this pressure depends on whether the gas is non-relativistic, relativistic, ultra-relativistic.
- non-relativistic, when particle velocities are much smaller than speed of light, pressure describes relation P = 2/3 u, where u means kinetic energy density.
- relativistic, when however particles approach the speed of light or are massless, energy and momentum relation becomes linear, and relativistic pressure takes form P = 1/3 u.
- ultra-relativistic means that particle velocity is extremely close to speed of light, takes form P = 1/3 u (where u is energy density).
Degeneracy pressure of neutrons however has its limit of effectiveness.
Above a certain critical mass (TOV limit) degenerated neutron gas ceases to be able to balance gravity, and equilibrium is broken.
(Even this quantum resistance has its limits. When a neutron star exceeds mass about 2.6 solar masses, degeneracy pressure is no longer enough. Further collapse begins nothing already stops gravity?.)
After crossing this limit nothing already stops further collapse of the core?.

PART 2 - MY HYPOTHESIS
And what happens next, when (Neutron Star) is even heavier than 2.6 solar masses?
Standard physics says: it collapses into --black hole with singularity - a point of infinite density (black hole) ....
I propose something different. Instead of a point - a structure. Instead of infinity - layers.

It's like peeling an onion: each layer is a different energy level, a different kind of particle, different physics.
In this work I show that inside (black hole?) there may exist a core with radius 15-48 meters, composed of four layers, which correspond to known particles from the Standard Model.
This is a proposal of a new look at what is hidden in the heart of (black hole?).
We go further wanting to describe what happens in further crushing of matter by gravity, which the neutron star could not stop.
The object that is formed after (Neutron Star), when collapse is not stopped is (black hole?).
We reached the limit after which (Neutron Star) no longer copes with gravity. It collapses further, but according to my model - not into an infinitely small point, but into a structure with dimensions from 15 to 48 meters. This is more or less as much as a football field. And this structure has layers.
Mechanisms in background during further collapse.
Beta decay and quantum tunneling
Before we go further, it is worth stopping at two mechanisms from nuclear physics and quantum mechanics that act at the level of single particles throughout the entire duration of collapse.
Beta decay and quark transformations (nuclear physics)
Inside nucleons (hadrons) (teams of particles (quarks (u,d,d)-neutron and quarks (d,u,u)-proton) bound by strong interaction through gluons (glue). transformations of quarks occur:
Inside each neutron and proton there are smaller particles - quarks. In extreme conditions these quarks can change their identity. Neutron can turn into proton, and proton into neutron.
It's a bit like a puzzle - blocks (quarks) rearrange, creating new shapes.
Beta decay (lonely neutron without proton pair, decay time 15 min.)
quarks (u,d,d) -->> quarks (d,u,u)
changing neutron into proton, electron and electron antineutrino.
reverse beta decay (stars, quantum tunneling, neutron star)
quarks (d,u,u) --> quarks (u,d,d)
proton captures electron, turning into neutron and emitting electron neutrino.
d - down
u - up
These transformations show that proton and neutron are not particles "permanently" in extreme conditions they can transform into each other.
(dissolution of W± and Z0 bosons responsible for beta decays and quantum tunneling)
Neutron and proton are not immutable balls. In extreme conditions - for example inside a collapsing star - they can change into each other. It's like transformation of water into steam or ice - depending on conditions.
Quantum tunneling (quantum physics)
Classical physics predicts that a particle without sufficient energy is not able to overcome energy barrier - for example Coulomb barrier resulting from repulsion of like charges.
Quantum mechanics however allows the phenomenon of tunneling: particle (electron, proton, neutron) has non-zero probability of "penetrating" through such a barrier, despite classically not having energy for it. This phenomenon is responsible for many nuclear reactions occurring inside stars, also in conditions of extreme pressure of matter.
Imagine that you stand before a high wall. Classical physics says: you need a ladder to climb. But quantum mechanics says: you have a small chance that you simply "penetrate" through the wall to the other side, even without a ladder. This phenomenon is called tunneling. In neutron stars and black holes it happens all the time - particles "penetrate" through barriers that normally are insurmountable for them.
Neutron and proton in extreme conditions can transform into each other.
(dissolution of W± and Z0 bosons responsible for beta decays and quantum tunneling)
Fundamental interactions
Further compression of matter by gravity encounters resistance resulting not only from quantum degeneracy, but also from nature of fundamental interactions that govern behavior of elementary particles.
(When we compress matter more and more, we encounter resistance that does not come from heat or gas pressure. It comes from the very forces that govern the universe from how particles interact with each other.)
bosons are carriers of interactions.
Interaction. Carrier particle (boson). Mass. Range.
Electromagnetism - Photon (y) - 0 - Infinite

Strong - Gluon (g) - 0 - 1e-15 m (subatomic)
Weak - Bosons W±, Z0 - 80.4-91.2 GeV/c2 - 1e-18 m (subatomic)
W± = 80.4 GeV - boson (weak int.).
Z0 = 91.2 GeV - boson (weak int.).
H0 = 125.1 GeV - Higgs boson.
H = 246 GeV - vacuum expectation value (Higgs field).
There are four fundamental forces in the universe.
Two of them - electromagnetism and strong interaction - carry massless carrier particles (photon and gluon).
But weak interaction carries heavy particles - bosons W± and Z0.
Their mass makes this interaction work only at very small distances smaller than diameter of atomic nucleus.
For giving mass to carrier particles - including bosons W± and Z0 - is responsible Higgs field, with vacuum expectation value equal to 246 GeV, and associated Higgs boson with mass 125 GeV.
Where do particles get mass from? Imagine that the universe is filled with dense honey - this is Higgs field.
Particles that move through this honey "catch" it on themselves and become heavy. The more particle "wades" through field, the greater its mass.
Higgs boson is like a "cluster" of this honey - a condensed drop that shows that field exists.
When we compress matter to extreme limits, this honey "melts" particles lose mass and become light as light (photon).

Electroweak symmetry restoration
At energies and densities close to conditions prevailing a fraction of a second after Big Bang, Higgs field "melts" (restoration of symmetry) - particles lose rest mass.
Weak interaction and electromagnetism, normally separate, merge then into one, uniform electroweak force.
Quark-gluon plasma
With even further compression of matter bonds between quarks in protons and neutrons break, creating quark-lepton-gluon plasma - dense "soup" of almost free quarks, leptons and gluons.
In this state strong interaction loses its usual property of quark confinement, and particles behave like ultra-relativistic gas.
P = 1/3 u ultra-relativistic equation of state.

PART 3 - MY HYPOTHESIS - Thought.
In this work the problem based on searching for full quantum theory of gravity is touched in favor of analysis of behavior of matter of Standard Model at extreme energy densities that accompany collapse of core with mass (Tolman-Oppenheimer-Volkoff limit (TOV).
It is estimated most often at 2.2-2.6 Msun. Above, neutron star core collapses further.
For calculations: Our Object M = 2.8 x Msun (we assume that already such mass of matter collapses into black hole)
R_core = G * M * m0 / X
rho_core = M / R_core^3
p_core = 1/3 * rho_core * c2 = 1/3 * M / R_core^3 * c2
eta = p_core * R_core^4 / G * M2 = 1/3 * c2 * m0 / X
Structural parameter eta is completely independent of mass M and gravitational constant G.
This means resistance of matter against gravity does not depend on macroscopic scale, but is dictated by universal microscopic constants of Standard Model and quantum mechanics!.

Constants:
Our Object M = 2.8 x Msun = 5.5692 x 1e30 kg
G = 6.674e-11 m3 kg-1 s-2
M = 5.569e30 kg
m0 = 1.675e-27 kg
c = 3e8 m/s
E_W± = 80.38 GeV, E_Z0 = 91.19 GeV, E_H0 = 125.1 GeV, E_H = 246 GeV

TABLE OF ENERGY SCALES, DENSITY AND RELATIVISTIC PRESSURE FOR 4 LAYERS OF OBJECT CORE (For mass M = 2.8 Msun = 5.569e30 kg)
1) Boson W± - 80.38 GeV - 48.35 m - 4.9272e25 kg/m3 - 4.4345e42 J/m3 - 1.4782e42 Pa
Boson Z0 - 91.19 GeV - 42.62 m - 7.1937e25 kg/m3 - 6.4743e42 J/m3 - 2.1581e42 Pa

2) Higgs boson - 125.10 GeV - 31.06 m - 1.8586e26 kg/m3 - 1.6727e43 J/m3 - 5.5758e42 Pa
Field Higgs - 246.00 GeV - 15.80 m - 1.4120e27 kg/m3 - 1.2708e44 J/m3 - 4.2359e43 Pa
Comparison: Neutron star- 10-13 km - 1e17-1e18 kg/m3
Inside (black hole?) (for mass 2.8 solar masses) there are four layers. Radii from 48 meters (outer) to 15 meters (inner). Diameter from 31 to 96 meters - size of football field.
Density is from 10 million to 10 billion times greater than in neutron star!
Phase 1: layers of bosons (W±, Z0) associated with weak interaction (beta decay, quantum tunneling). Their melting by gravitational squeeze.
Phase 2: layers associated with Higgs field (H0, H), where mass ceases to be constant feature. 15 meter piece of space in which gravity restored state of Universe a fraction of second after Big Bang (about 1e-12 second), before Higgs field changed its state and gave particles mass.

Physics used: GTR, Nuclear, Standard Model, Quantum, Thermodynamics, Electrodynamics, Astrophysics, Classical/Relativistic Mechanics, Physical Mathematics.

Leptons: e- 0.511 MeV, nu_e <0.8 eV, mu- 105.7 MeV, nu_mu <0.19 MeV, tau- 1777 MeV, nu_tau <18.2 MeV
Quarks: u 2.2 MeV +2/3, d 4.7 MeV -1/3, c 1.27 GeV +2/3, s 93.4 MeV -1/3, t 172.7 GeV +2/3, b 4.18 GeV -1/3

GeV to Joules:
E_W± = 80.38 GeV = 80.38e9 x 1.602e-19 = 1.2877e-8 J -> R_core 48.35 m Diam 96.7 m
E_Z0 = 91.19 GeV = 1.4609e-8 J -> R_core 42.62 m Diam 85.2 m
E_H0 = 125.1 GeV = 2.0041e-8 J -> R_core 31.06 m Diam 62.1 m
E_H = 246 GeV = 3.9409e-8 J -> R_core 15.80 m Diam 31.6 m

SUBSTRATE / SUPPLEMENT - Calculations and Tables
Microscopic Quantum Mechanisms 2.8 Msun

Marcin Slowik - ORCID 0009-0004-6707-4546 - Zielona Gora, Poland
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Formula: R_core = G*M*m0 / X, rho = M / R_core^3, p = 1/3 rho c^2, eta = 1/3 c^2 m0 / X independent of M,G

Constants: G=6.674e-11, M=2.8*1.989e30=5.5692e30 kg, m0=1.675e-27 kg, c=3e8, 1eV=1.602e-19 J, W± 80.38 GeV, Z0 91.19 GeV, H0 125.1 GeV, H 246 GeV

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W± 80.38 GeV = 1.2877e-8 J -> 48.35 m (96.7 m diam) rho 4.9272e25 p 1.4782e42
Z0 91.19 GeV = 1.4609e-8 J -> 42.62 m (85.2 m) rho 7.1937e25 p 2.1581e42
H0 125.1 GeV = 2.0041e-8 J -> 31.06 m (62.1 m) rho 1.8586e26 p 5.5758e42
H 246 GeV = 3.9409e-8 J -> 15.80 m (31.6 m) rho 1.4120e27 p 4.2359e43
Leptons: e- 0.511 MeV, nu_e <0.8 eV, mu- 105.7 MeV, nu_mu <0.19 MeV, tau- 1777 MeV, nu_tau <18.2 MeV
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